

Chapter 9: Linear momentum and collisions

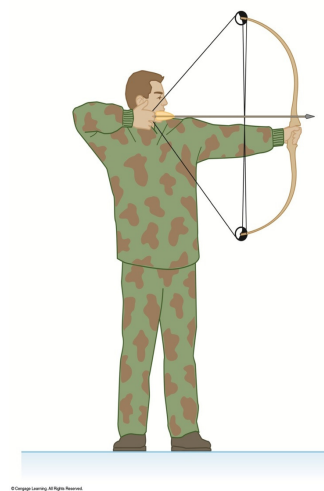
Linear momentum
Isolated system
Non- Isolated system

Collisions
Centre of mass
Rockets

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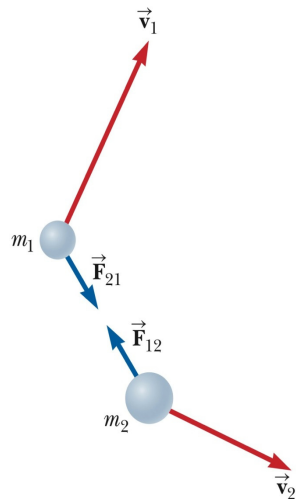
Linear Momentum

- ▶ An archer stands on ice with his bow and arrow.
- ▶ What will happen if he shoots the arrow?
- ▶ Newton's 3rd law
- ▶ Kinematics – acceleration?
- ▶ Forces
- ▶ Energy
- ▶ Linear momentum can offer an easy solution.



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Linear Momentum



- ▶ Consider an isolated two particle system.
- ▶ No external forces
- ▶ only $\vec{F}_{12}$ en $\vec{F}_{21}$
- ▶ Masses m_1, m_2
- ▶ Velocities $\vec{v}_1, \vec{v}_2$

$$\vec{F}_{12} = -\vec{F}_{21}$$

Newton 3

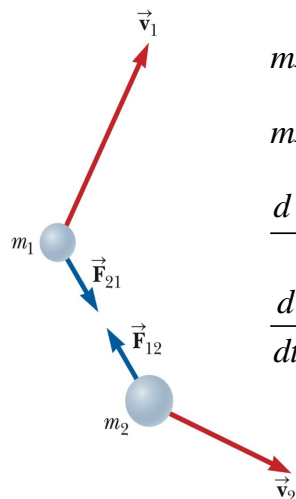
$$\vec{F}_{12} + \vec{F}_{21} = 0$$

$$m_1 \vec{a}_{12} + m_2 \vec{a}_{21} = 0$$

Newton 2

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Linear Momentum



$$m_1 \vec{a}_1 + m_2 \vec{a}_2 = 0$$

$$m_1 \frac{d\vec{v}_1}{dt} + m_2 \frac{d\vec{v}_2}{dt} = 0$$

$$\frac{d(m_1 \vec{v}_1)}{dt} + \frac{d(m_2 \vec{v}_2)}{dt} = 0 \quad \text{If } m_1, m_2 \text{ are constant}$$

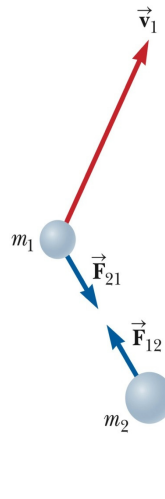
$$\frac{d}{dt}(m_1 \vec{v}_1 + m_2 \vec{v}_2) = 0$$

$$\therefore m_1 \vec{v}_1 + m_2 \vec{v}_2 \quad \text{Must be constant}$$

$\vec{p} = m\vec{v}$ This is known as the linear momentum of an object that can be considered as a particle.

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Linear Momentum



- ▶ $\vec{p} = m\vec{v}$
- ▶ $\vec{p}$, $\vec{v}$ are in the same direction
- ▶ In general $\vec{p}$ will have 3 components
- ▶ $\vec{p}_x = m\vec{v}_x$, $\vec{p}_y = m\vec{v}_y$, $\vec{p}_z = m\vec{v}_z$
- ▶ For a 2-particle system
- ▶ $\vec{p}_1 = m_1\vec{v}_1$, $\vec{p}_2 = m_2\vec{v}_2$
- and
- $\vec{p}_{total} = \vec{p}_1 + \vec{p}_2$

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Linear Momentum – isolated system

▶ $\frac{d}{dt}(m_1\vec{v}_1 + m_2\vec{v}_2) = 0$

$\frac{d}{dt}(\vec{p}_1 + \vec{p}_2) = \frac{d}{dt}(\vec{p}_{total}) = 0$

$\vec{p}_{total} = \text{constant}$

▶ $\Delta\vec{p}_{total} = 0$ in the interval Δt or dt

$\Rightarrow \vec{p}_{tot,f} - \vec{p}_{tot,i} = 0$

$\vec{p}_{tot,f} = \vec{p}_{tot,i}$

▶ $\vec{p}_{i,1} + \vec{p}_{i,2} = \vec{p}_{f,1} + \vec{p}_{f,2}$

2 -particle system

Conservation of linear momentum

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Linear Momentum

$$\begin{aligned} \triangleright p_{i,1x} + p_{i,2x} &= p_{f,1x} + p_{f,2x} \\ p_{i,1y} + p_{i,2y} &= p_{f,1y} + p_{f,2y} \\ p_{i,1z} + p_{i,2z} &= p_{f,1z} + p_{f,2z} \end{aligned}$$

- ▶ 3 components in the conservation law
- ▶ If no external forces act on the system (isolated), linear momentum will be conserved.

$$\Delta \vec{p}_{total} = 0$$

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Linear Momentum

- ▶ We can determine the relationship between the net force $\vec{F}$ that acts on a particle and the particle's linear momentum

- ▶ For a single particle, from Newton's 2nd law

$$\begin{aligned} \triangleright \sum \vec{F} &= m\vec{a} = m \frac{d\vec{v}}{dt} \\ &= \frac{d(m\vec{v})}{dt} \quad \text{If } m \text{ is constant} \end{aligned}$$

$$\sum \vec{F} = \frac{d\vec{p}}{dt}$$

- ▶ This is the general form of Newton's 2nd law.

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Linear Momentum

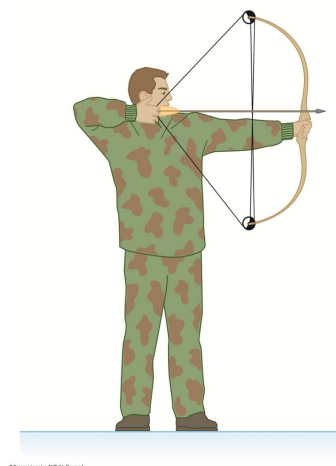
- If m is **not constant**:

$$\begin{aligned}\sum \vec{F} &= \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt} \\ &= \frac{dm}{dt} \vec{v} + m \frac{d\vec{v}}{dt} \\ \sum \vec{F} &= \frac{dm}{dt} \vec{v} + m\vec{a}\end{aligned}$$

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Example

- A 60 kg archer stands at rest on frictionless ice and fires a 0.030 kg arrow horizontally at 85 m/s. With what velocity does the archer move across the ice after firing the arrow?



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Example

- ▶ A 60 kg archer stands at rest on frictionless ice and fires a 0.030 kg arrow horizontally at 85 m/s. With what velocity does the archer move across the ice after firing the arrow?
- ▶ System: Archer + Bow, arrow
- ▶ Gravitational force and normal force are present – non-isolated
- ▶ But these forces are not horizontal.
- ▶ No external forces in the horizontal direction – therefore we can consider this as an isolated system.

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Non-isolated system

- ▶ Net force on the system is not zero.
- ▶ The net force acts on the system in a time interval Δt .
- ▶ Momentum can be transferred to the system from the environment.
- ▶ Consider that the net force varies with time, then

$$\sum \vec{F} = \frac{d\vec{p}}{dt}$$

$$d\vec{p} = \sum \vec{F} dt$$

- ▶ We integrate this equation

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Non-isolated system

- ▶ Integrating

$$\int_{p_i}^{p_f} d\vec{p} = \int_{t_i}^{t_f} \sum \vec{F} dt$$

$$\vec{p}|_{p_i}^{p_f} = \int_{t_i}^{t_f} \sum \vec{F} dt$$

$$p_f - p_i = \Delta\vec{p} = \int_{t_i}^{t_f} \sum \vec{F} dt$$

- ▶ To determine the integral on the right hand side, we need to know what the net force looks like.

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Non-isolated system

- ▶ $p_f - p_i = \Delta\vec{p} = \int_{t_i}^{t_f} \sum \vec{F} dt$

- ▶ We define the left-hand side as the **impulse**

- ▶ $\vec{I} = \int_{t_i}^{t_f} \sum \vec{F} dt$

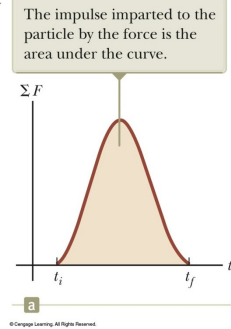
- ▶ The impulse gives an **indication of** the extent to which the net force **changes the momentum**.

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Non-isolated system

- ▶ Net force acts over an interval $\Delta t = t_f - t_i$.
- ▶ A graph of force vs time
- ▶ Impulse has a magnitude equal to the area under the curve over the interval Δt .
- ▶ The direction of the impulse, $\vec{I}$, is the same as $\Delta \vec{p}$.
- ▶ The units are the same as for momentum: kg.m.s^{-1} .

$$\vec{I} = \int_{t_i}^{t_f} \sum \vec{F} dt$$



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Non-isolated system

- ▶ Since the net force varies with time, the average net force can be defined as

$$\left(\sum \vec{F} \right)_{avg} = \frac{1}{\Delta t} \int_{t_i}^{t_f} \sum \vec{F} dt$$

- ▶ where $\Delta t = t_f - t_i$

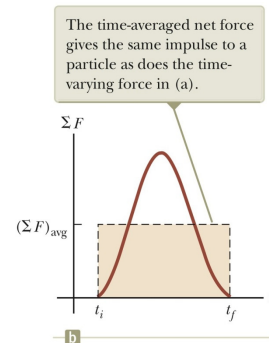
- ▶ And therefore

$$\vec{I} = \left(\sum \vec{F} \right)_{avg} \Delta t$$

- ▶ If the force is constant

$$\left(\sum \vec{F} \right)_{avg} = \sum \vec{F}$$

$$\text{en } \vec{I} = \sum \vec{F} \Delta t$$



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Non-isolated system

- ▶ The change in momentum is equal to the impulse of the net force that acts on the particle

$$\Delta \vec{p} = \vec{I}$$

- ▶ This is the general form of the momentum conservation law.
- ▶ In an isolated system

$$\begin{aligned} \sum \vec{F} = 0 &\Rightarrow \vec{I} = 0 \\ &\Rightarrow \Delta \vec{p} = 0 \end{aligned}$$

- ▶ p_i and p_f are the momentum just before and just after the collision.

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Non-isolated system

- ▶ **IMPULSE-APPROXIMATION:**
Assume that one force acts for a short period of time on the particle.
- ▶ The net force $\sum \vec{F} \approx \vec{F}$, the single force, since all the other forces are much smaller than the one force, we can ignore the other forces, such as gravitation on the particle.
- ▶ Example: Ball is hit by a bat
- ▶ The forces involved are a few thousand Newton.

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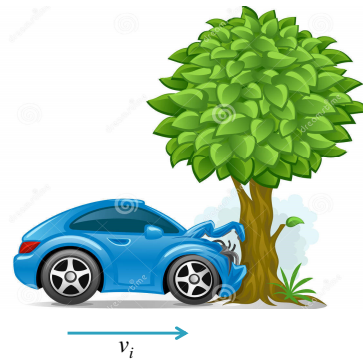
Example

A car moves at 20 m/s and collides with a tree. Find the magnitude of the average force that acts on the passenger who has a mass of 65 kg in the following situations:

- a) The passenger does not use a safety belt. He comes to rest against the windscreen, and the collision lasts 3.0 ms.
- b) The car has an airbag. The passenger is stopped by the force that acts on the passenger for 30 ms.

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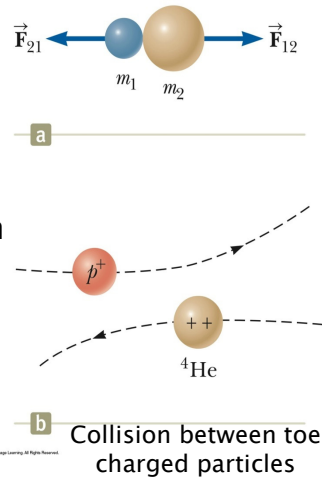
Example (a)



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Collision in 1-D

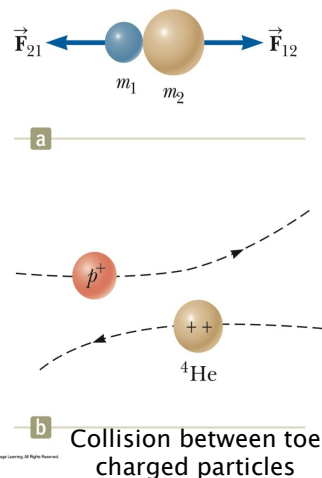
- ▶ A collision represents an event during which two particles come close to each other and interact by means of forces.
- ▶ The internal forces can be complex and can vary with time, but at the instant of the collision the internal forces are much larger than any of the other external forces.



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Collisions in 1-D

- ▶ Types of collisions
- ▶ **Elastic collisions**
 - Linear momentum is conserved
 - No change in the kinetic energy of the system
- ▶ **Inelastic collisions**
 - Linear momentum is conserved, but kinetic energy is not conserved.
 - E.g. Rubber ball collides with the floor.
 - Perfectly inelastic collision
 - Two objects collide and stick together after the collision – move as one object.

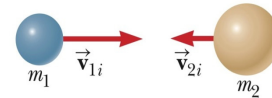


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Collision in 1-D: Perfectly inelastic collision

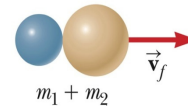
- ▶ $\Delta \vec{p}_{tot} = 0$
- ▶ $\vec{p}_{tot,f} - \vec{p}_{tot,i} = 0$
- ▶ $m_1 \vec{v}_{1,i} + m_2 \vec{v}_{2,i} = (m_1 + m_2) \vec{v}_f$
- ▶ $\vec{v}_f = \frac{m_1 \vec{v}_{1,i} + m_2 \vec{v}_{2,i}}{m_1 + m_2}$
- ▶ Special case: $v_{2,i} = 0$, m_2 is initially at rest
- ▶ $\vec{v}_f = \frac{m_1}{m_1 + m_2} \vec{v}_{1,i}$
- ▶ $m_1 \gg m_2 \Rightarrow \vec{v}_f \approx \vec{v}_{1,i}$
- ▶ $m_1 \ll m_2 \Rightarrow \vec{v}_f \approx 0$

Before the collision, the particles move separately.



a

After the collision, the particles move together.



b

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Collision in 1-D: elastic collision

- ▶ The linear momentum is conserved, and the kinetic energy is conserved.

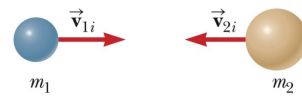
$$\vec{p}_{tot,i} = \vec{p}_{tot,f}$$

$$m_1 \vec{v}_{1,i} + m_2 \vec{v}_{2,i} = m_1 \vec{v}_{1,f} + m_2 \vec{v}_{2,f} \quad (1)$$

$$K_i = K_f$$

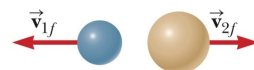
$$\frac{1}{2} m_1 \vec{v}_{1,i}^2 + \frac{1}{2} m_2 \vec{v}_{2,i}^2 = \frac{1}{2} m_1 \vec{v}_{1,f}^2 + \frac{1}{2} m_2 \vec{v}_{2,f}^2 \quad (2)$$

Before the collision, the particles move separately.



a

After the collision, the particles continue to move separately with new velocities.



b

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Collision in 1-D: elastic collision

- From ②

$$m_1 \vec{v}_{1,i}^2 - m_1 \vec{v}_{1,f}^2 = +m_2 \vec{v}_{2,f}^2 - m_2 \vec{v}_{2,i}^2$$

$$m_1 (\vec{v}_{1,i}^2 - \vec{v}_{1,f}^2) = +m_2 (\vec{v}_{2,f}^2 - \vec{v}_{2,i}^2)$$

$$m_1 (\vec{v}_{1,i} - \vec{v}_{1,f})(\vec{v}_{1,i} + \vec{v}_{1,f}) = +m_2 (\vec{v}_{2,f} + \vec{v}_{2,i})(\vec{v}_{2,f} - \vec{v}_{2,i}) \quad ③$$

- From ①

$$m_1 \vec{v}_{1,i} - m_1 \vec{v}_{1,f} = +m_2 \vec{v}_{2,f} - m_2 \vec{v}_{2,i}$$

$$m_1 (\vec{v}_{1,i} - \vec{v}_{1,f}) = +m_2 (\vec{v}_{2,f} - \vec{v}_{2,i}) \quad ④$$

- ③ / ④

$$(\vec{v}_{1,i} + \vec{v}_{1,f}) = (\vec{v}_{2,f} + \vec{v}_{2,i})$$

$$\Rightarrow \vec{v}_{1,i} - \vec{v}_{2,i} = -(\vec{v}_{1,f} - \vec{v}_{2,f}) \quad ⑤$$

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Collision in 1-D: elastic collision

- From ⑤

$$\vec{v}_{2,f} = \vec{v}_{1,i} - \vec{v}_{2,i} + \vec{v}_{1,f} \quad ⑥$$

- From ⑥ in ①

$$m_1 \vec{v}_{1,i} + m_2 \vec{v}_{2,i} = m_1 \vec{v}_{1,f} + m_2 (\vec{v}_{1,i} - \vec{v}_{2,i} + \vec{v}_{1,f})$$

$$(m_1 - m_2) \vec{v}_{1,i} + 2m_2 \vec{v}_{2,i} = (m_1 + m_2) \vec{v}_{1,f}$$

$$\vec{v}_{1,f} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) \vec{v}_{1,i} + \left(\frac{2m_2}{m_1 + m_2} \right) \vec{v}_{2,i} \quad ⑦$$

- From ⑤ for $\vec{v}_{1,f}$ in ①

$$\vec{v}_{2,f} = \left(\frac{2m_1}{m_1 + m_2} \right) \vec{v}_{1,i} + \left(\frac{m_2 - m_1}{m_1 + m_2} \right) \vec{v}_{2,i} \quad ⑧$$

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Collision in 1-D: elastic collision

Special cases

- ▶ Special case $m_1 = m_2$

- ▶ From ⑦ and ⑧

$$\vec{v}_{1,f} = \vec{v}_{2,i}$$

$$\vec{v}_{2,f} = \vec{v}_{1,i}$$

- ▶ Special case $v_{2i} = 0$

- ▶ From ⑦ and ⑧

$$\vec{v}_{1,f} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) \vec{v}_{1,i}$$

$$\vec{v}_{2,f} = \left(\frac{2m_1}{m_1 + m_2} \right) \vec{v}_{1,i}$$

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Collision in 1-D: elastic collision

Special cases

- ▶ Special case $m_1 \gg m_2$ and $v_{2i} = 0$

$$\vec{v}_{1,f} \approx \vec{v}_{1,i}$$

$$\vec{v}_{2,f} \approx 2\vec{v}_{1,i}$$

- ▶ Example: Uranium ion colliding with hydrogen ion

- ▶ Special case $m_2 \gg m_1$ and $v_{2i} = 0$

$$\vec{v}_{1,f} \approx -\vec{v}_{1,i}$$

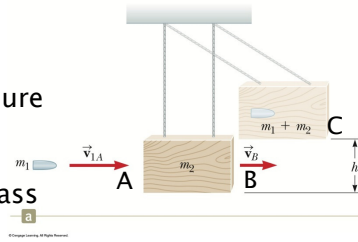
$$\vec{v}_{2,f} \approx 0$$

- ▶ Example: Tennis ball colliding with bowling ball

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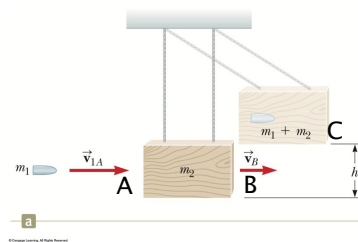
Example 9.6

The ballistic pendulum is used to measure the speed of fast moving projectiles such as bullets. A projectile of mass m_1 is fired into a large wooden block of mass m_2 that is suspended from two light wires. The projectile embeds in the block and the entire system swings through a height h . Determine the speed of the projectile.



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Example 9.6



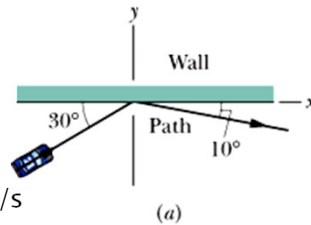
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Example

The figure shows an overhead view of the path taken by a race car driver as his car collides with the track wall. Just before the collision he is travelling at speed $v_i = 70 \text{ m/s}$ along a straight line at 30° from the wall. Just after the collision he is travelling at speed $v_f = 50 \text{ m/s}$ along a straight line at 10° from the wall. His mass $m = 80 \text{ kg}$.

(a) What is the impulse on the driver due to the collision?

(b) If the collision lasts for 14 ms , what is the magnitude of the average force on the driver during the collision?



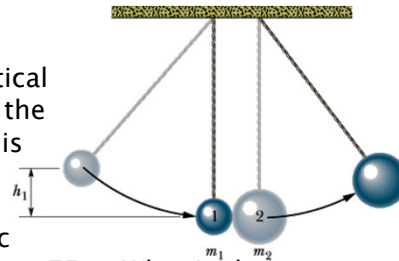
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Example

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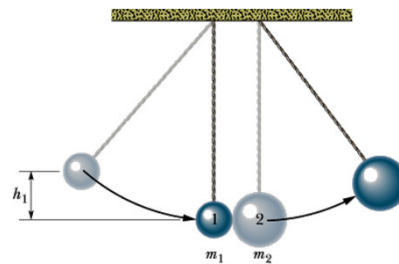
Example

Two metal spheres, suspended by vertical cords, initially just touch, as shown in the figure. Sphere 1, with mass $m_1 = 30 \text{ g}$ is pulled to the left to a height $h_1 = 8.0 \text{ cm}$ and then released from rest. After swinging down, it undergoes an elastic collision with sphere 2, whose mass $m_2 = 75 \text{ g}$. What is the velocity v_{1f} of sphere 1 just after the collision?



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Example



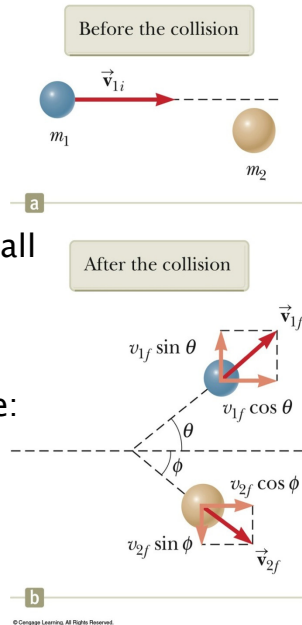
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Collisions in 2-D

- ▶ Momentum is conserved in all collisions
- ▶ If the collision is in 2- or 3-D the momentum must be conserved in all 2 (or 3) dimensions.
- ▶ When two balls move in 2 dimensions, and collide, the components of the momentum are:

$$m_1 \vec{v}_{1,ix} + m_2 \vec{v}_{2,ix} = m_1 \vec{v}_{1,fx} + m_2 \vec{v}_{2,fx}$$

$$m_1 \vec{v}_{1,iy} + m_2 \vec{v}_{2,iy} = m_1 \vec{v}_{1,fy} + m_2 \vec{v}_{2,fy}$$
- ▶ Note that the velocity now has 3 subscripts



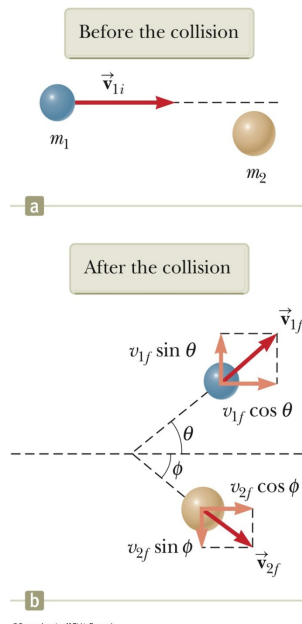
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Collisions in 2-D

- ▶ In the specific case of a glancing collision:
- ▶ The 2 objects move in different directions after the collision.
- ▶ m_1 moves at an angle θ , and m_2 moves at an angle ϕ both to the original direction
- ▶ $\Delta p_x = 0 \rightarrow p_{ix} = p_{fx}$

$$\rightarrow m_1 v_{1i} = m_1 v_{1f} \cos \theta + m_2 v_{2f} \cos \phi$$
- ▶ $\Delta p_y = 0 \rightarrow p_{iy} = p_{fy}$

$$\rightarrow 0 = m_1 v_{1f} \sin \theta - m_2 v_{2f} \sin \phi$$



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Collisions in 2-D

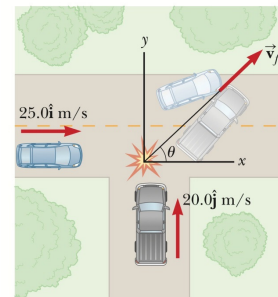
- ▶ If the collision is elastic, then the KE is also conserved:

$$K_i = K_f$$
- ▶
$$\frac{1}{2}m_1\vec{v}_{1,i}^2 = \frac{1}{2}m_1\vec{v}_{1,f}^2 + \frac{1}{2}m_2\vec{v}_{2,f}^2$$
- ▶ If the collision is not elastic we cannot use the KE conservation.

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Example 9.8

A 1500 kg car travelling east with a speed of 25.0 m/s collides at an intersection with a 2500 kg truck travelling north at a speed of 20.0 m/s. Find the direction and magnitude of the velocity of the wreckage after the collision, assuming the vehicles stick together after the collision.



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Example 9.8

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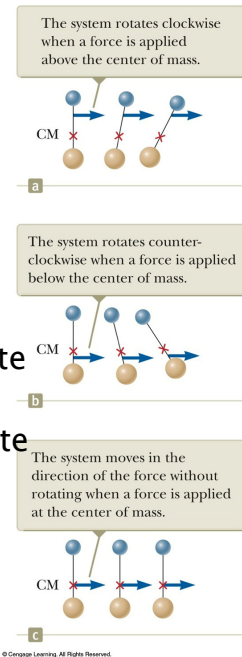
Example 9.9

A proton collides elastically with another proton that is initially at rest. The incoming proton makes a glancing collision with the second one and moves off at an angle of 37° to the original direction, while the second proton moves off at an angle ϕ , relative to the same direction.

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Centre of mass

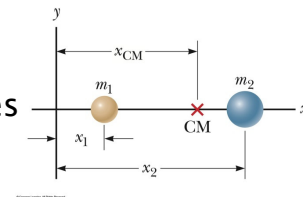
- ▶ The centre of mass – CM, com
- ▶ The com of a system of particles can be described as the **average position of the system's mass**.
- ▶ If a force is applied:
 - ▶ (a) above the com, the system will rotate clockwise.
 - ▶ (b) below the com, the system will rotate anti-clockwise.
 - ▶ (c) at the com, the system will move in the direction of the force and not rotate.



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Centre of mass

- ▶ The com of a two particle system lies somewhere between the particles.
- ▶ $x_{CM} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$
- ▶ If $x_1 = 0$, $x_2 = d$, $m_2 = 2m_1$,
 $x_{cm} = 2/3d$ cm lies closer to m_2 .
- ▶ If the two masses are the same,
 x_{cm} is midway between the two.
- ▶ We can extend the system to n number of particles:



$$x_{CM} = \frac{m_1 x_1 + m_2 x_2 + \dots + m_n x_n}{m_1 + m_2 + \dots + m_n} = \frac{\sum_i m_i x_i}{\sum_i m_i} = \frac{1}{M} \sum_i m_i x_i$$

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Centre of mass

- ▶ If the particles are distributed in 2 or 3 dimensions:

$$x_{CM} = \frac{1}{M} \sum_i m_i x_i \quad y_{CM} = \frac{1}{M} \sum_i m_i y_i \quad z_{CM} = \frac{1}{M} \sum_i m_i z_i$$

- ▶ The position of the com is

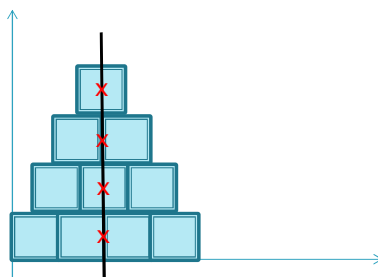
$$\begin{aligned} \vec{r}_{CM} &= x_{CM} \hat{i} + y_{CM} \hat{j} + z_{CM} \hat{k} \\ &= \frac{1}{M} \sum_i m_i x_i \hat{i} + \frac{1}{M} \sum_i m_i y_i \hat{j} + \frac{1}{M} \sum_i m_i z_i \hat{k} \end{aligned}$$

$$\vec{r}_{CM} = \frac{1}{M} \sum_i m_i \vec{r}_i$$

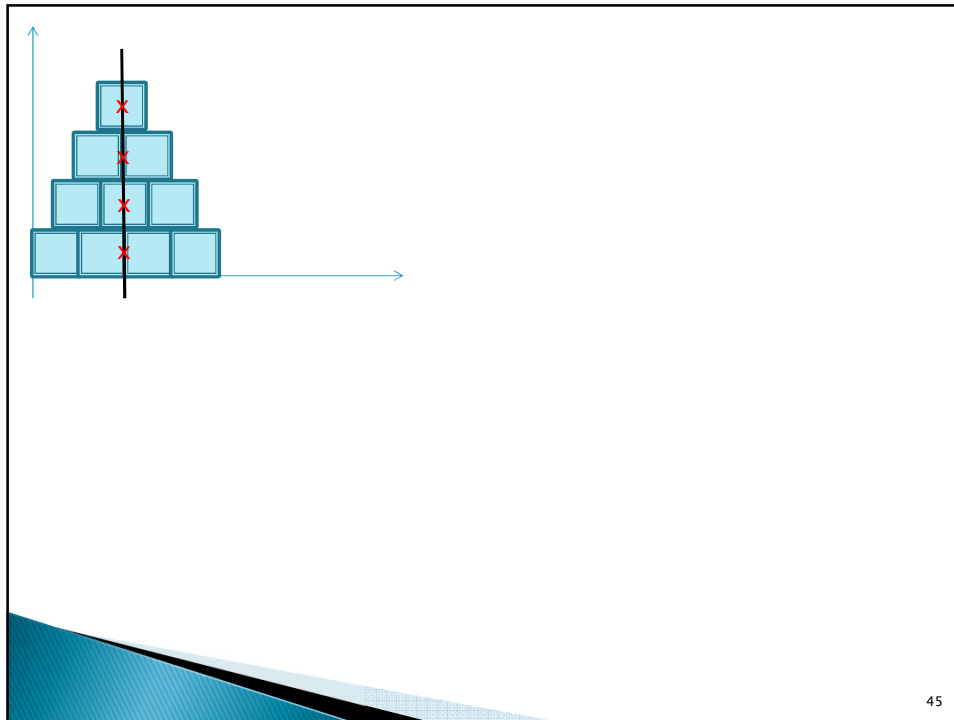
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Example

- ▶ We pack some blocks which are 5 cm wide and 5 cm high on top of each other as shown in the figure. Each block has a mass m . The total mass of the system is $\sum m_i = 10m$
- ▶ Determine the i_{com} of the system.



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System of particles

- ▶ System of 2 or more particles
- ▶ Dynamics (movement) of the CM can be determined
- ▶ We first determine the velocity and acceleration of the CM

$$\vec{v}_{cm} = \frac{d\vec{r}_{cm}}{dt} = \frac{1}{M} \sum_i m_i \frac{d\vec{r}_i}{dt} = \frac{1}{M} \sum_i m_i \vec{v}_i$$

- ▶ $\vec{v}_{cm} = \frac{1}{M} \sum_i m_i \vec{v}_i$ $\vec{v}_i$ is the velocity of each particle

$$\vec{a}_{cm} = \frac{d\vec{v}_{cm}}{dt} = \frac{1}{M} \sum_i m_i \frac{d\vec{v}_i}{dt} = \frac{1}{M} \sum_i m_i \vec{a}_i$$

- ▶ $\vec{a}_{cm} = \frac{1}{M} \sum_i m_i \vec{a}_i$ $\vec{a}_i$ is the acceleration of each particle

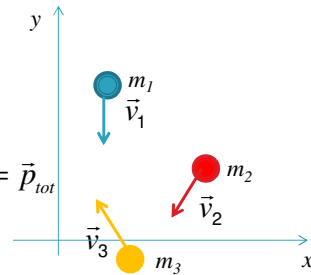
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System of particles

► Momentum

$$M\vec{v}_{cm} = M \left(\frac{1}{M} \sum_i m_i \vec{v}_i \right) = \sum_i m_i \vec{v}_i = \sum_i \vec{p}_i = \vec{p}_{tot}$$

$$\vec{p}_{tot} = M\vec{v}_{cm}$$



► Force

$$M\vec{a}_{cm} = M \left(\frac{1}{M} \sum_i m_i \vec{a}_i \right) = \sum_i m_i \vec{a}_i = \sum_i \vec{F}_i$$

$$\sum_i \vec{F}_i = \sum \vec{F}_{ext}$$

$$\sum \vec{F}_{ext} = M\vec{a}_{cm}$$

F_i can include internal and external forces. Internal forces will add up to zero. Only external forces remain.

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System of particles

$$\int \sum \vec{F}_{ext} dt = \int M\vec{a}_{cm} dt$$

$$= M \int \vec{a}_{cm} dt$$

$$= M \int \frac{d\vec{v}_{cm}}{dt} dt$$

$$\int \sum \vec{F}_{ext} dt = M \int_{\vec{v}_{cm,i}}^{\vec{v}_{cm,f}} d\vec{v}_{cm}$$

$$\int \sum \vec{F}_{ext} dt = M \Delta \vec{v}_{cm} = M (\vec{v}_{cm,f} - \vec{v}_{cm,i})$$

$$\vec{I} = \Delta \vec{p}_{tot}$$

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System of particles

- ▶ For an isolated system

$$\vec{F}_{ext} = 0$$

$$\Rightarrow \vec{I} = 0$$

$$\Delta \vec{p}_{tot} = 0$$

$$\vec{p}_{tot} = \text{konstante}$$

- ▶ Linear momentum conservation law

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Example

- ▶ A 2000 kg Mercedes car just started moving away from a stop sign, when it was hit from behind by a 1000 kg Volkswagen. During the collision the bumpers hook on each other and the two cars slide over a distance of 3 m before coming to a stop. The driver of the Mercedes states that she was moving at 5 m/s when the collision occurred. Did the driver of the Volkswagen exceed the speed limit of 60 km/h? Assume that there is friction between the tyres and the road, with $\mu_k = 0.8$.

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Example

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